

Julian Mueller

(+1) 248-862-1220 | julianmu@umich.edu | linkedin.com/in/julianmue/ | github.com/mujulian | julianmueller.xyz

Education

University of Michigan - Ann Arbor

Bachelor of Science in Computer Science

Ann Arbor, MI

Aug. 2021 – May 2025

Professional Experience

Control Systems Engineer

August 2025 – Present

Thermo Systems

- Expanded a **C++ telemetry ingestion server** to support two additional cooling device types contributing to over **\$400 million in yearly operational cost savings for our client**
- Built an automated reporting service using **Django, Python, ReportLab, and MySQL**, leveraging the Django ORM to optimize relational queries and **saving our team 800+ hours in manual reporting**
- Optimized legacy AVEVA VBA code by implementing faster, modern, string functions, **speeding up field testing by 60%**
- Delivered software and automation solutions for enterprise clients, translating requirements into control logic and graphics

Quality Systems Intern

September 2024 – August 2024

Manga Steyr

- Engineered an auditing web app (JavaScript, HTML, CSS), **increasing user capacity by 660%** and **cutting \$300 in monthly costs** by replacing legacy software
- Built a part validity scanner that **identified 500+ defective parts**, improving vendor quality and reducing downstream assembly issues.
- Prepared and presented quality systems data in daily stand up meetings and weekly client meetings

Projects

JGram | *React, Python, Flask, SQLite3, AWS*

- Developed a scalable full-stack Instagram clone using React and Flask with real-time updates, including user auth, image posts, and live interactions.
- Leveraged React.js to render dynamic content, implemented features such as user authentication, pagination, likes, and comments with real-time updates
- Engineered authentication, data retrieval, and user interaction via Flask in a scalable and efficient manner

Paxos ShardKV Database | *Go, Bash*

- Developed a distributed key-value database in Go using the Paxos consensus algorithm, enabling fault-tolerant replication and consistent state synchronization across multiple server replicas.
- Designed and optimized Paxos communication workflows to reduce unnecessary consensus phases, improving agreement efficiency and lowering coordination overhead.

Storyverse | *Python, TypeScript, LangChain, OpenAPI, FastAPI*

- Developed a RAG (Retrieval Augmented Generation) system to read and fine tune story prompts in order to deliver age appropriate stories to readers
- Led development of a responsive React frontend (Chakra UI, React Hooks) for an AI-based educational tool, improving usability for **100+ student testers**
- Optimized page rendering and component efficiency using React hooks and Chakra UI, enhancing the overall user experience

Technical Skills

Languages: C/C++, Python, Go, JavaScript, TypeScript, SQL, HTML, CSS

Developer Tools: Git/Github, Unix/Linux, VS Code, Visual Studio, GoLand, Jira, Amazon Web Services (AWS), Docker, VirtualBox, Access, Excel

Frameworks: React.js, Express.js, Spring Boot, Flask, FastAPI, OpenAPI, LangChain, Jinja2, Bootstrap, NumPy, Chakra UI, pandas, BeautifulSoup, Selenium, Matplotlib, Coverage.py

Concepts: Software Engineering, Full Stack, Frontend, Backend, Software Quality Assurance, Distributed Systems, Web Frameworks, Agile/Scrum, REST API